

Cognitive Performance and Environmental Exposure: A Multi-Level Data Analysis Using NHANES, EPA, and BRFSS Data

CS 210 Data Management for Data Science Final Project Report

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Github Link: <https://github.com/hemadharshinii-s/cognitive-performance-project>

Demo Video:

<https://drive.google.com/drive/folders/1uvVcpg-FWgw5queYnIEzMg-lLj1wUz0h?usp=sharing>

Problem Definition & Motivation

Cognitive performance, a measure of how efficiently the brain processes information (and accounts for mental abilities like memory, attention, reasoning, and decision-making), is influenced by a variety of factors, including demographic, biological, behavioral, and environmental factors. As a student majoring in Cognitive Science (along with Computer Science and Data Science), I wanted to explore the factors that affect cognitive performance and determine the extent to which these effects are. Through my project, I investigated whether cognitive performance can be predicted and explained by combining demographic factors (age, gender, education), health indicators (body mass index), behavioral variables (smoking, physical activity), and environmental exposure (air quality).

The central question explored is:

“To what extent can cognitive performance be predicted using demographic, health, behavioral, and environmental variables?”

My planned approach (as I outlined in my project proposal) did not work out as intended, and I ended up exploring this question through:

1. Individual-Level Prediction (NHANES)
2. Population-Level Environmental Comparison (NHANES & EPA AQI)
3. External Validation Using BRFSS & EPA AQI

Thus, rather than being a single unified modeling framework, my project combines predictive modeling and exploratory analysis across datasets. Even though each notebook is separate, they form a multi-level framework for examining cognitive performance from individual, population, and external validation perspectives.

Throughout this project, I applied various core concepts from this course, including data cleaning and preprocessing, data integration, and predictive modeling. I imported pandas and numpy libraries to help me read the datasets and efficiently work with the data. I cleaned, transformed, and merged multiple real-world datasets using structured pipelines and SQL-based joins on unique identifiers. When needed, I performed feature engineering, categorical encoding, and data normalization to help prepare data for analysis. I implemented machine learning models, after importing the relevant libraries, to perform regression and classification tasks. All technical aspects of this project are accompanied by my own analysis and interpretations.

Research, Novelty, & Importance

Cognitive decline is commonly associated with aging, but research suggests that cognition is also influenced by socioeconomic status, lifestyle behaviors, and environmental conditions.

Key established findings from my research include:

- Age is strongly negatively associated with cognitive performance (which is consistent with the established understanding of cognitive aging)
- Higher education is associated with better cognitive performance
- Smoking and inactivity are linked to poorer cognitive health
- Air pollution exposure is associated with neuroinflammation and cognitive decline

My project builds on these findings by using nationally representative datasets (NHANES & BRFSS) and environmental data (EPA's AQI data) to examine whether these relationships can be observed empirically at both individual and population levels.

This research is important because cognitive decline has major implications for aging populations, healthcare systems, and quality of life, and environmental factors (like air pollution) are relevant in public health research and discussions. This project explores whether cognitive outcomes can be meaningfully linked to both personal and external factors, and the findings contribute to a broader understanding of how demographic, behavioral, and environmental variables interact.

Dataset Descriptions

National Health and Nutrition Examination Survey (NHANES)

The National Health and Nutrition Examination Survey (NHANES), a CDC Research Program, contains datasets providing in-depth individual-level health and demographic information. The datasets I used for my analysis include: Demographics (DEMO_H), Cognitive Function (CFQ_G & CFQ_H), Body Measures (BMX_H), Smoking Behavior (SMQ_H), and Physical Activity (PAQ_H). From each dataset, I used only columns containing relevant information for my study, specifically those that could be considered features/predictors of cognitive performance. These features include age, gender, education level, height, weight, body mass index, lifetime smoking status, and physical activity. As a measurement of cognitive performance, I utilize the Digit Symbol Substitution Test score from the CFQ datasets. Because NHANES contains cognitive performance data only for 2011-2014, the datasets I collected are from 2013-2014, except for CFQ_G, which is from 2011-2012. All datasets were merged using a column that contained a unique ID for each participant.

Environmental Protection Agency (EPA)

The Environmental Protection Agency (EPA) is an independent federal agency tasked with protecting human health and the environment, and they monitor various environmental factors, including air quality. For my project, I used the EPA's Air Quality Index (AQI) data as a proxy for environmental exposure. Air Quality Index is a numerical scale used to report daily air quality, highlighting how clean or polluted the air is (lower AQIs indicate better air quality). To align with NHANES data, I collected EPA AQI data at the county level for 2011-2014. To align with BRFSS data, I collected EPA AQI data for 2022.

Behavioral Risk Factor Surveillance System (BRFSS)

The Behavioral Risk Factor Surveillance System (BRFSS) is a premier system, run by the CDC and state health departments, of health-related telephone surveys that collect data from U.S. residents on risk behaviors, chronic conditions, and preventive services. I used this dataset for analyzing population-level cognitive impairment. Specifically, the features I kept were age group, education, BMI, smoking status, and diabetes status. Here, the measure of cognitive performance was based on a binary self-reported score of cognitive decline. The BRFSS had no datasets available for 2013-2014, so I used a 2022 dataset instead.

Data Limitations

- NHANES cognitive data only includes participants aged 60+, with age top-coded at 80 for confidentiality, so after merging, the dataset was reduced to roughly 1500 participant entries
- BRFSS cognitive measure is self-reported, so it is not a clinical assessment
- Cross-sectional design of the datasets when merged limits interpretations to associations only, and no causal inference should be made between features

Methodology & Pipeline

During production, my project evolved iteratively due to data availability constraints and my adaptations of my original proposal to align with real-world dataset limitations. So after many iterations, I've structured my project as a multi-stage data analysis pipeline across three complementary notebooks. Each notebook addresses the research question at a different level of analysis: individual-level prediction, population-level trend comparison, and external validation.

01 Predicting Cognitive Performance Using Demographic, Health, and Behavioral Factors (NHANES 2013-2014)

This notebook focuses on predicting cognitive performance using individual-level demographic, health, and behavioral variables. While my initial plan was to integrate AQI data at the state level here, NHANES does not provide state-level geographic identifiers for public use, so I decided to only utilize NHANES datasets and opted out of exploring environmental features for now.

METHODOLOGY

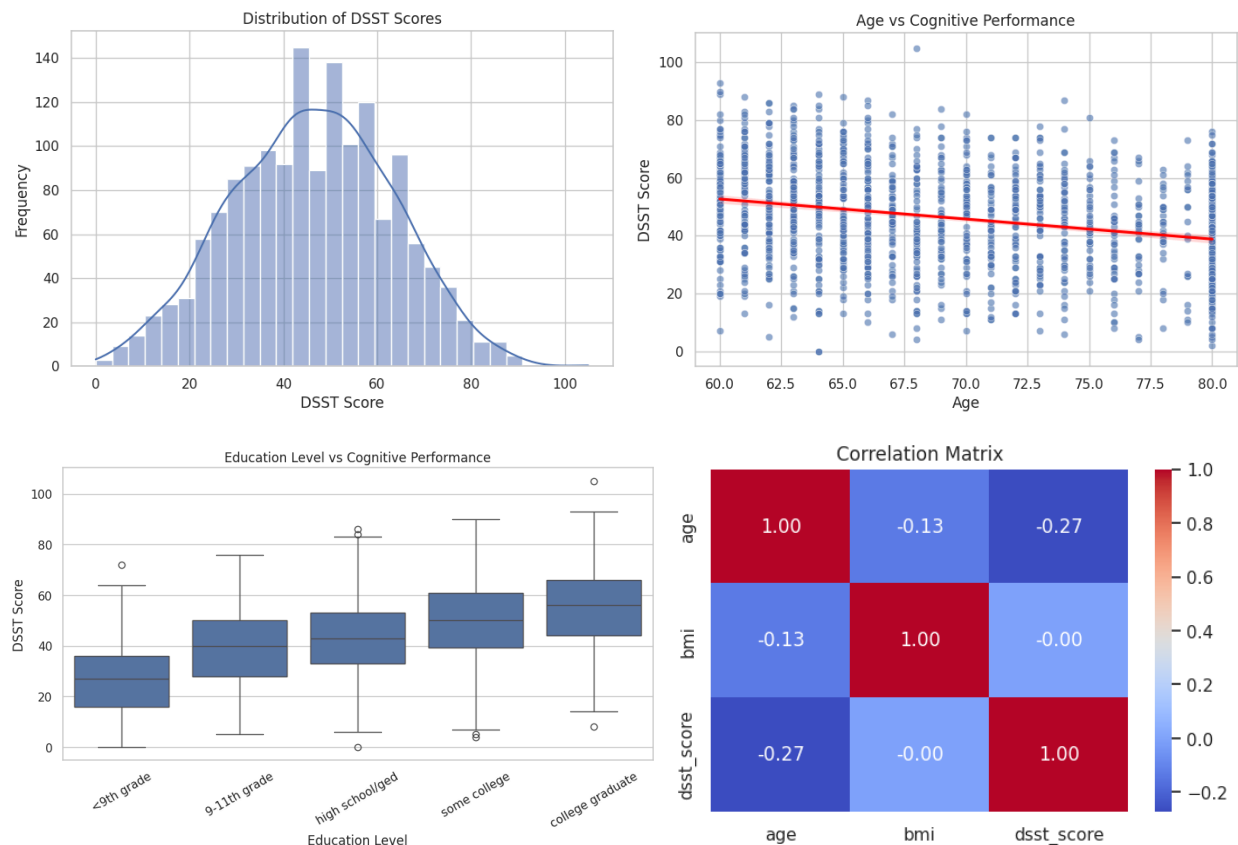
My analysis follows a structured pipeline:

1. Data Acquisition & Integration: I loaded multiple NHANES datasets from 2013-2014 (listed in the Dataset Descriptions), to extract features related to Demographics, Cognitive Function, Body Measures, Smoking Behavior, and Physical Activity. I used SQL inner joins (which also effectively handled all missing variables, resulting in each participant's entry containing no missing features) to merge the datasets on a unique participant identifier (SEQN).
2. Data Cleaning & Feature Engineering: I selected a few variables from each dataset, removing irrelevant variables to reduce dimensionality and keeping variables that may impact cognitive performance. I standardized and mapped categorical variables (like gender, education, smoking status, and physical activity) to readable labels, and used height and weight info from the Body Measures dataset to compute Body Mass Index.

3. Modeling Approach: I trained and tested three models—a Linear Regression model, a Decision Tree Regressor, and a Random Forest Regressor. The data was split into training (80%) and testing (20%) sets, and one-hot encoding was applied to categorical variables using a preprocessing pipeline.
 - a. Target Variable: I assessed cognitive performance using the Digit Symbol Substitution Test (DSST), which provides a continuous score ranging from 0-105, using this score as my target variable.
 - b. I evaluated the accuracy of the models using Root Mean Square Error (RMSE, which captured prediction error magnitude) and R^2 (a measure that indicates how much of the variation of the target feature is explained by independent features in the regression model).

VISUALIZATIONS

Before performing any machine learning, I used visualizations to explore the merged dataset and to look for any patterns or correlations between various features and cognitive performance (by DSST score). Interpretations and analysis of these visualizations are offered in the next section.



RESULTS & INTERPRETATIONS

Comparing the accuracies of each model, the Linear Regression model achieved the best predictive performance with an R^2 of approximately 0.39, outperforming both of the tree-based models. This R^2 score is actually relatively meaningful in this context as outcomes are influenced by many unobserved and noisy factors. It suggests that the model captures a non-trivial portion of variability in cognitive performance, given the complexity of human behavior and the limited feature set I've used in modeling. However, this does not necessarily strictly imply that the underlying relationship between the features and target variable is purely linear. Rather it may indicate that the available features contain limited non-linear structure, or that more complex interactions were not fully captured due to constraints in the dataset and feature engineering process.

Model	R^2	RMSE
Linear Regression	0.394648	13.359189
Decision Tree	0.337413	13.976471
Random Forest	0.337274	13.977939

Key Observations from the Visualizations & Training:

- Age showed a slight negative association with DSST score, which may confirm popular research supporting the claim that cognitive decline is expected with aging
- Education shows itself to be a strong predictor of cognitive performance, which may indicate a strong socioeconomic influence on cognitive performance
- BMI, smoking status, and physical activity showed weak predictive power individually (not much correlation between each feature and cognitive performance by itself)
- Evaluating the non-linear models (Decision Tree and Random Forest) did not significantly improve performance, suggesting limited complex interactions in this feature set

Overall, the Linear Regression model explains a moderate portion of variance in cognitive performance, but a substantial amount remains unexplained, which may indicate additional factors that I did not measure/include in this analysis and modeling.

02 Population-Level Comparison of Air Quality and Cognitive Performance (2011-2014)

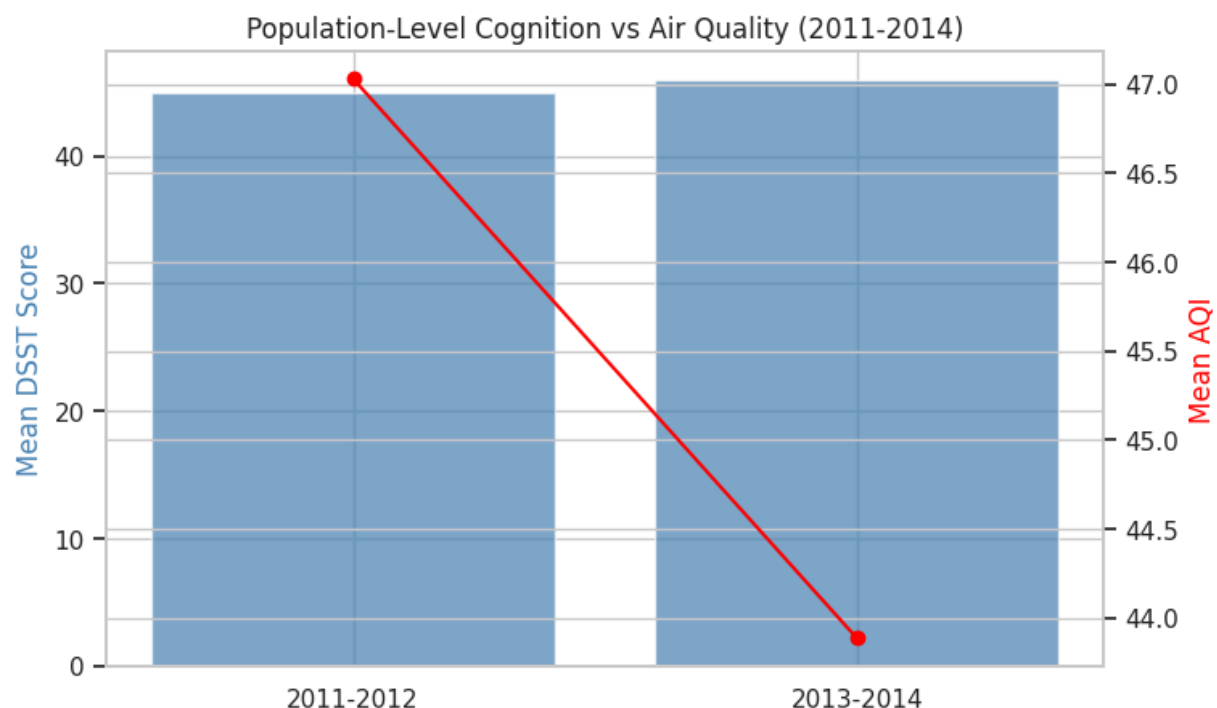
This notebook shifts the analysis from individual-level prediction to population-level trend comparison between cognitive performance and environmental exposure. Since I couldn't merge NHANES and AQI datasets due to public restriction on geographic identifiers in the NHANES dataset, I performed this exploratory analysis to see if I could identify any correlation between cognitive performance and air quality index. I do note that this analysis absolutely does not indicate any causal relationship between the two factors.

METHODOLOGY

My pipeline for this exploration consists of:

1. Data Preparation: I utilized NHANES DSST data from the 2011-2012 and 2013-2014 cycles, and EPA AQI data from 2011, 2012, 2013, and 2014, aggregated by year.
2. Aggregation: I aggregated NHANES DSST scores to compute mean cognitive performance per cycle and aggregated AQI values to compute national averages. I then aligned the two, to get two cycles with an average cognitive performance and average national AQI each.
3. Merging: I merged the cycle-level cognitive and environmental summaries into a single dataset for comparison.

VISUALIZATION: Interpretation and analysis of this visualization is offered in the next section



RESULTS & INTERPRETATIONS

As seen in the plot, across the two NHANES cycles:

- Mean DSST scores increased very slightly (just barely) from the 2011-2012 cycle to the 2013-2014 cycle
- Mean AQI decreased over the same period, indicating improved air quality

This suggests a weak inverse relationship between air pollution and cognitive performance at the population level (mean AQI decreased while mean DSST score increased—in context, the average air quality improved while average cognitive performance improved).

However, the magnitude of this change is small and the analysis is very limited, as I've only used two time points and used independent cross-sectional samples. There is also a lack of control for demographic shifts across cycles.

Therefore the observed trend is *descriptive only* and does not support causal inference.

03 External Validation: Air Quality and Cognitive Impairment Across U.S. States (BRFSS + EPA AQI 2022)

This final notebook evaluates whether environmental exposure (AQI) is associated with cognitive impairment at the state level, using an independent dataset. Since I was unable to pursue my original plan of combining NHANES and EPA data at the state level, I looked for another dataset that could somewhat make up for it, to still assess whether air quality has any effects on cognitive performance. It is important to note, however, that the measure for cognitive performance here is a binary self-reported perception of cognitive difficulty, which is not clinical by any means and may be heavily subjective.

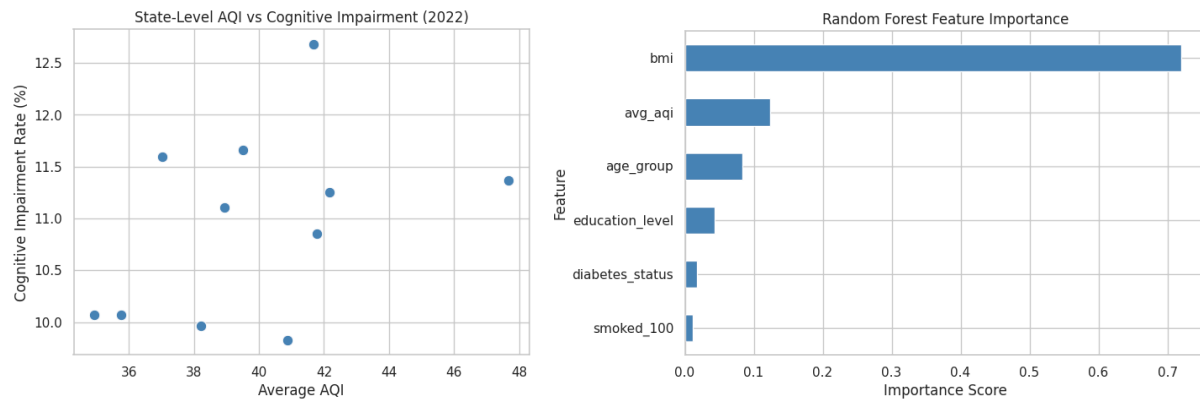
METHODOLOGY

The pipeline I used for this notebook:

1. **Data Processing:** I filtered the BRFSS survey dataset to keep relevant cognitive and health variables (to serve as features). Cognitive outcome was defined as a binary variable where 1 indicated self-reported cognitive difficulty and 0 indicated no difficulty.
2. **Geographic Mapping:** The dataset utilized State FIPS codes rather than state names, so I referenced the U.S. Census website to map these codes to state names to align with EPA AQI data. It's important to acknowledge that the cleaned dataset only contains 10 states, which significantly limits generalizability, so any observed patterns should be interpreted as exploratory and not representative of national-level trends.
3. **Environmental Data Integration:** I then aggregated AQI values from the EPA to state-level averages for the year of 2022 (to align with the 2022 BRFSS dataset). I merged both datasets together using state names.
4. **Modeling Approach:** Since the target variable is binary rather than continuous, I utilized classification models (instead of the regression models I used in the first notebook). I trained and tested three models—a Logistic Regression model, a Decision Tree Classifier, and a Random Forest Classifier. I imputed any missing values using median imputation. I imported `accuracy_score` from `sklearn` to evaluate the accuracy of the models. Initially, I received very high accuracy scores for all three models, causing me to check the dataset balance in the target variable. Noticing a significant imbalance between classes, I added a balanced class weight to models in hopes of receiving a more accurate reflection of model performance. With this adjustment, the accuracy score was supplemented with precision, recall, and F1-score for better understanding of model performance.

VISUALIZATIONS

I used visualizations to better explore and understand relationships between the data (especially between AQI and cognitive impairment). Interpretations and analysis of these visualizations are offered in the next section.



RESULTS & INTERPRETATIONS

Initially, I was surprised to see that model performance was consistently high across all three classification models. With the Logistic Regression model performing the best with ~88.6% accuracy, followed by the Decision Tree model with ~88.5% accuracy, and finally the Random Forest model with ~85.1% accuracy. However, upon inspection, it seemed very likely that these accuracy values were heavily inflated by class imbalance, as most respondents reported no cognitive impairment (i.e., there isn't an even split between those who self-reported impairment and those who self-reported no impairment).

To adjust for this, I set each model's class weights to balanced (applying `class_weight = "balanced"`), after which overall accuracy decreased unevenly across models (the logistic regression and decision tree models experienced substantial drops in accuracy, while the random forest model remained relatively stable). However, the classification report was more informative of model performance and revealed a tradeoff between accuracy and minor-class detection within each model. Now, the Random Forest model maintained the highest overall accuracy (~84.2%), but it performed poorly in identifying cognitive impairment cases (recall, as provided in the classification report, was only 0.07). This indicates a strong bias towards the majority class. The Logistic Regression model however achieved a more balanced performance, with a much higher recall for the minority class (0.56), suggesting better sensitivity (recall) to cognitive impairment. Finally, the Decision Tree model showed the highest recall (0.67) but lower precision, indicating increased false positives. With a dataset like this, where there is class imbalance, accuracy score as an evaluation metric can be misleading because it mainly reflects majority-class performance. For us, recall and class-wise metrics provided a more meaningful evaluation of model behavior.

Summary of Model Performance Before and After Applying Class Weighting

Model	Accuracy (before balancing)	Accuracy (after balancing)	Behavior
Logistic Regression	0.8856	0.5922	balanced detection (best for minority class)
Decision Tree	0.8855	0.5194	high recall, unstable precision
Random Forest	0.8509	0.8419	high accuracy, but poor minority detection

Key Observations from the Visualizations & Training:

- BMI emerged as the strongest predictor in the Random Forest model (which is surprising as, in the NHANES datasets, there didn't appear to be a strong correlation between BMI and cognitive performance)
- AQI showed some predictive importance, suggesting a potential environmental influence on cognitive performance
- Education level (which seemed to have a decent/significant/obvious correlation with cognitive performance in the NHANES dataset) contributed less than expected in this dataset

Overall, it's possible that AQI contributes some predictive value, but individual health indicators appear to remain the stronger determinants of reported cognitive impairment.

Overall Analysis

Though all three datasets explored the same question in different ways, there were some consistent patterns or strong observations (limited to these datasets and analyses):

- Age (negative relationship) and education level (positive relationship) appear to be consistent predictors of cognition
- BMI and AQI show themselves as moderate or inconsistent predictors across different datasets
- Smoking status, physical activity, and diabetes (for the BRFSS model exclusively) were all weak predictors by themselves

Drawing insight from each of the three notebooks: the individual-level modeling showed that demographic variables dominate predictive power; population-level analysis suggested weak but potential relationships between air quality and cognition; and external validation confirms that environmental exposures may contribute modestly/somewhat to cognitive performance, but is likely not the primary driver of reported cognitive outcomes.

TAKEAWAY: Cognitive performance appears to be heavily influenced by demographic and socioeconomic factors (e.g., education level), with environmental exposure (specific to air quality in our analysis) playing a secondary but potentially meaningful role at the population level.

Summary of Challenges & Limitations

As mentioned throughout this report, I faced several data and methodological constraints which shaped the development and direction of my projection. To reiterate:

- NHANES only contains cognitive performance/function data and scores for participants aged 60+ while age is also top-coded for confidentiality at age 80. This limits generalizability as findings cannot be representative of younger populations and may be biased toward aging-related cognitive patterns
- NHANES does not provide detailed geographic identifiers, which prevented me from directly merging its datasets with EPA Air Quality Data and from analyzing environmental and cognitive data at the individual level, instead requiring separate, aggregated comparisons
- The datasets I used are not necessarily aligned (related to the same time period) (due to dataset availability constraints), as I focus the first two notebooks on the years 2011-2014 and focus the third notebook on 2022, which limits direct comparability across the three levels of analyses
- The measurement of cognitive performance I used differs across datasets and analyses. While NHANES uses a clinical DSST score, BRFSS relies on self-reported cognitive difficulty, which is more subjective and less precise
- All analyses are cross-sectional and it must be noted that these results reflect associations rather than causal relationships

Conclusion

All-in-all, this project investigated cognitive performance using a multi-level data approach combining NHANES, EPA air quality data, and BRFSS survey data. Across all analyses, the findings suggest that:

- Cognitive performance can be partially predicted using demographic and health variables, with moderate accuracy (which aligns with what I expected)
- Environmental exposure (air quality) maintains weak but observable associations with cognitive outcomes at the population level (which slightly surprised me, as I expected air quality to have a bit more impact due to its neurological effects)

The results support the conclusion that cognitive health is multifactorial, where demographic and socioeconomic variables contribute strongly in comparison to environmental exposure alone (this is again, specific to the datasets analyzed in this project).

Future Work

I can think of several extensions that could be made to improve and expand this analysis (that I hope to explore further in the future), including: incorporating additional NHANES variables like income and diet; using geographically matched individual-level pollution exposure rather than aggregated AQI at the national or state level; better addressing the class imbalance in BRFSS using resampling or cost-sensitive learning methods; and exploring more advanced predictive models such as gradient boosting models.